Point-to-Point Response to Reviewers

Journal: *China Communications* — **Manuscript ID**: CNCOMM-2026-0222

Title: *System Relaxation for Resilient Network Governance: A Model-Driven Control Mechanism Against Structural Stress Concentration*

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Dear Associate Editor-in-Chief Prof. Ying-Chang Liang, Associate Editor, and Anonymous Reviewers,

We would like to express our deepest gratitude to the Associate Editor-in-Chief, the Associate Editor, and the two anonymous Reviewers for dedicating your valuable time and expertise to reviewing our manuscript (ID: **CNCOMM-2026-0222**). We are deeply grateful for the thoroughness of your review and sincerely appreciate every critique and suggestion, which exposed important mathematical imprecisions, unstated assumptions, and empirical gaps in our original submission.

In accordance with the journal guidelines, all revisions in the main manuscript (`main_ccjnl.pdf` / `main.pdf`) are **highlighted in blue text**, and the complete step-by-step mathematical proofs (Sections S-A through S-J) are provided in the accompanying standalone **Supplementary Material (`sm.pdf`)** so that the main manuscript remains concise and well-structured. Below, we respond to **every single comment individually**—first restating the Editor's or Reviewer's suggestion, and then presenting our humble response and exact manuscript changes.

Part I: Individual Responses to Associate Editor (Editor: 2)

Associate Editor — Comment AE.1 (Establishing Technical Novelty, Problem Formulation, and Systematic Comparison)

Reviewer's Suggestion (Restated): The Associate Editor advises that the technical novelty of SRA should be more clearly established by explicitly distinguishing it from existing adaptive routing and load-balancing approaches—especially those based on dynamic link weights and centrality optimization—and by providing a clearer mathematical problem formulation along with a systematic comparison against closely related methods.

Our Humble Response: We sincerely thank the Associate Editor for this insightful guidance. We humbly acknowledge that in our original manuscript, the problem was introduced at a relatively high level, and the boundary between SRA and classical traffic engineering methods—such as Interior Gateway Protocol (IGP) dynamic weight optimization and static centrality-aware routing—was not articulated clearly enough. Following your advice, we have substantially revised the manuscript in four aspects:

1. **Explicit Research Gap and Structured 6-Column Taxonomy (Section I, Table I, and Section II-B)**: We added a dedicated *Research Gap* discussion in Section I and expanded **Table I** into a feature-by-feature comparison across five paradigms: *Reactive Queue-Aware Routing*, *IGP Link-Weight Optimization (Fortz & Thorup [26])*, *Open-Loop Static Centrality Routing [19]*, *Data-Driven AI/DRL Routing [21, 22]*, and *Proposed SRA*. We clarify that while reactive queue-based methods act only after physical buffer buildup (causing phase lag) and online IGP weight tuning requires full traffic matrices and induces severe route flapping (98.3% switching frequency when run online), SRA couples a continuous leaky-integrator structural stress state $S_i^{(k)}$ with minimum dwell-time block gating T_d to proactively reshape routing costs without chattering.

2. **Formal Mathematical Problem Formulation (Section III-A, Eq. (5))**: We formulated the resilient network governance task as a constrained discrete-time bi-criteria optimization problem (**Eq. (5)**) that minimizes long-run peak generalized structural stress $\limsup_{k\to\infty} \max_{i\in V} \widetilde{\kappa}_i(S^{(k)}) + \eta_{\mathrm{sw}} \bar{f}_{\mathrm{sw}}$ subject to leaky-integrator state dynamics $S^{(k+1)} = (1-\alpha)S^{(k)} + \alpha q^{(k)}$, linear cost actuation $C^{(k)} = \mathbf{1} + \beta_I S^{(k)}$, and minimum dwell-time switching spacing $\tau_{m+1} - \tau_m \geq T_d$.

3. **Direct Experimental Benchmarking Against Closely Related Methods (Section IV-F, Table 4 / Table IV, and Fig. 10)**: We implemented both *Online IGP Weight Tuning (Fortz–Thorup)* and *Static Betweenness Centrality Routing* alongside *LIRP-D* and *Dijkstra* across 30 independent random seeds on BA scale-free networks ($N = 100, \rho = 0.9$). As shown in **Table 4 (Table IV)**, Full SRA achieves top-tier throughput (89.76 ± 0.76 pkts/step, 99.65% PDR) while cutting control-plane route switching frequency by 79.7% (20.0% vs. 98.3% under online IGP weight tuning).

Associate Editor — Comment AE.2 (Theoretical Rigor in Theorems 1 & 2, Lemma 2, and Proposition 3)

Reviewer's Suggestion (Restated): The Associate Editor recommends carefully re-examining the convergence claims in Theorems 1 and 2—particularly the distinction between attraction to a limit set and convergence to a unique equilibrium—and providing stronger mathematical support for the relationship between structural stress reduction and network capacity improvement in Lemma 2 and Proposition 3.

Our Humble Response: We are deeply grateful to the Associate Editor for pointing out these critical theoretical issues. Upon careful re-examination, we humbly admit that our original phrasing in Theorem 1 conflated global set attraction with convergence to a unique fixed point, and that the proofs of Theorem 2, Lemma 2, and Proposition 3 contained mathematical gaps (including a reversed minimum inequality in the original Appendix J). We have thoroughly rewritten **Sections III-C, III-D, and III-E** as well as **Supplementary Material Sections S-A, S-B, S-C, S-D, S-I, and S-J**:

1. **Distinction Between Limit Set Attraction (Ω) and Unique Point Equilibrium (Theorem 1, Remark 2, and Supplement Sections S-A–S-C)**: We first completed the coordinate-wise box-invariance proof in **Lemma 1 (Supplement Section S-A)** by explicitly defining $\delta_{k,i}^{+} := \max\{S_i^{(k)} - 1, 0\}$ and $\delta_{k,i}^{-} := \max\{-S_i^{(k)}, 0\}$. Next, using the Krasovskii regularization $\mathcal{N}(S) := \bigcap_{\delta>0} \overline{\mathrm{co}}(\mathrm{norm}(\kappa(B_\delta(S))))$ and Aubin–Cellina upper-semicontinuous set-valued map theory [28], we revised **Theorem 1** to establish global attraction on $X = [0,1]^N$ to the non-empty, compact, positively invariant limit set $\Omega = \bigcap_{n\geq 0} F^n(X) \subset X$ ($\lim_{k\to\infty} \mathrm{dist}(S^{(k)}, \Omega) = 0$). We also added **Remark 2** to explicitly explain that Ω is a compact attracting set (which may contain bounded limit cycles near multi-path tie-breaking boundaries) rather than a single static point equilibrium, and that unique fixed-point convergence holds conditionally within a stable routing stratum (**Corollary 1**).

2. **Exact Discrete-Time Drop-Jump Energy Bound (Theorem 2 and Supplement Section S-D)**: We removed the heuristic continuous-time analogy and derived an exact discrete-time one-step difference inequality for the implemented normalized feedback $q^{(k)} = \mathrm{norm}(\kappa^{(k)}) \in [0,1]^N$ and tracking energy $E_k = \|S^{(k)} - q^{(k)}\|_2^2$, proving via Weighted Young's inequality ($\lambda = \alpha/(1-\alpha)$) that $E_{k+1} \leq (1-\alpha)E_k + \frac{1}{\alpha}\|q^{(k)} - q^{(k+1)}\|_2^2 \leq (1-\alpha)E_k + \bar{\Delta}^2/\alpha$ and $\limsup_{k\to\infty} E_k \leq \bar{\Delta}^2/\alpha^2$.

3. **Non-Circular Peak-Shaving Certificate (Lemma 2 and Supplement Section S-I)**:

We replaced the circular conservation argument with an explicit **Route-Redistribution Budget** $B_j^{(m)} := (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+$ and **Non-Overloading Diversion Condition** ($\kappa_j^{(m)} + B_j^{(m)} \leq \kappa_{\max}^{(m)} - \delta$ for all $j \neq u$), proving a rigorous sufficient condition under which path switching strictly reduces global peak stress ($\kappa_{\max}^{(m+1)} \leq \kappa_{\max}^{(m)} - \delta$).

4. **Rigorous Capacity Bridge (Proposition 3 and Supplement Section S-J)**: We corrected the reversed inequality to $\rho_{\max}(\kappa) = \min_{i \in V}(\mu_i/\kappa_i) \geq \mu_{\min}/\kappa_{\max} =: L(\kappa)$ and proved three exact results: (i) unconditional expansion of the worst-case guaranteed capacity lower bound $L(\kappa') > L(\kappa)$; (ii) exact inverse proportionality $\rho_{\max}(\kappa')/\rho_{\max}(\kappa) = \kappa_{\max}/\kappa'_{\max} > 1$ under homogeneous capacities ($\mu_i \equiv \mu$) or $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max} > 1$ under generalized capacity-normalized stress $\widetilde{\kappa}_i(S; W, \mu)$; and (iii) the exact joint (`AND`) sufficient service-rate slack condition ($\mu_u/(\kappa_{\max} - d) \geq \rho_0$ and $\mu_j/(\kappa_j + B_j) \geq \rho_0,\ \forall j \neq u$) under heterogeneous capacities with unweighted centrality.

Associate Editor — Comment AE.3 (Strengthening Experimental Evaluation: Ablations, Sensitivity, Baselines, Statistics, Heterogeneous/Time-Varying Traffic, and Complexity/Runtime)

Reviewer's Suggestion (Restated): The Associate Editor encourages strengthening the experimental evaluation by including comprehensive ablation studies, parameter sensitivity analysis, stronger baselines, statistical validation across multiple trials, experiments under heterogeneous and time-varying traffic conditions, and evaluations of computational complexity and runtime overhead.

Our Humble Response: We gratefully accept this suggestion and humbly agree that our initial empirical evaluation needed broader baselines, statistical confidence intervals, and non-stationary/real-world testbeds. We have conducted a comprehensive multi-seed simulation campaign (30 independent trials per configuration, reporting mean ± SD and normal-approximation 95% confidence intervals $[\bar{x} \pm 1.96\, s/\sqrt{30}]$) and added three new subsections (**Sections IV-F, IV-G, and IV-H**), two new quantitative tables (**Tables 4 and 5 / Tables IV and V**), and six multi-panel figures (**Figs. 6, 7, 10, 11, 12, and 13**):

1. **Component Ablation and Stronger Baselines (Section IV-F, Table 4 / Table IV, and Fig. 10)**: Isolates each SRA module (*Memoryless Reactive BC* $\alpha = 1$, *Unnormalized SRA*, *SRA w/o Dwell* $T_d = 1$, and *Full SRA*) against *Dijkstra*, *LIRP-D*, *Static BC*, and *Online IGP Weight Tuning (Fortz–Thorup)*.

2. **Heterogeneous Capacities, Skewed Pareto Gravity Demand, Time-Varying Surge, and Telemetry Noise (Section IV-G and Fig. 11)**: Evaluates SRA under heavy-tailed Pareto gravity demands ($w_i \sim$ Pareto(1.5)), heterogeneous node capacities, a non-stationary traffic surge, and multiplicative sensing noise $\sigma \in \{0\%, 5\%, 10\%, 20\%\}$.

3. **Hyperparameter Sensitivity, Scalability ($N = 50..500$), Real Carrier/AS Backbones (`Internet AS`, `Deltacom`, `UsCarrier`), and Runtime Benchmarks (Section IV-B, Section IV-H, Table 5 / Table V, Figs. 12–13)**: Restores the 3D (α, β_I) sensitivity surface (**Fig. 12**), evaluates dwell-time sensitivity $T_d \in \{1..10\}$ (**Fig. 13(a)**), and benchmarks wall-clock runtime and throughput scaling on real-world carrier/AS backbones (`Internet AS` $N = 60$, +18.91% throughput gain; `Deltacom` $N = 113$, +52.24% gain; `UsCarrier` $N = 158$, +81.07% gain) as well as synthetic graphs up to $N = 500$ (902.95 ms/eval, +108.33% relative throughput gain).

Associate Editor — Comment AE.4 (Underlying Assumptions, Limitations of Betweenness Centrality, Presentation, and Reproducibility)

Reviewer's Suggestion (Restated): The Associate Editor asks the authors to clarify the underlying model assumptions, discuss the limitations of using betweenness centrality as a stress indicator, and improve overall presentation and reproducibility.

Our Humble Response: We thank the Associate Editor for emphasizing scientific transparency and reproducibility. We have addressed each aspect as follows:

1. **Explicit Model Assumptions (Section III-A)**: We formalized **Assumptions 1–4** covering 2-vertex-connected alternative paths, time-scale separation between control epochs and packet forwarding, deterministic tie-breaking/ECMP, and quasi-stationary demand/capacity parameters.
2. **Limitations of Unweighted Betweenness Centrality & Generalized Formulation (Section III-A Eq. (2), Section IV-G, and Section V-B)**: We explicitly discuss why unweighted topological BC can mismatch actual queuing load under skewed traffic or heterogeneous capacities, introduce the demand- and capacity-weighted stress $\widetilde{\kappa}_i(S; W, \mu)$ in **Eq. (2)**, and consolidate topological failure cases (homogeneous regular lattices and single-bridge cut-vertex topologies) and scalability limitations in a dedicated subsection (**Section V-B**).
3. **Presentation & Open-Source Reproducibility (Section V-C)**: We refined the typesetting of the manuscript, moved lengthy proofs into the standalone **Supplementary Material (`sm.pdf`)**, and publicly released all simulation code, raw multi-seed outputs, and LaTeX sources at https://github.com/unicorners-compin/SRA-Resilient-Network-Governance.

Part II: Individual Responses to Reviewer 1 (Comments R1.1 – R1.34)

We sincerely thank Reviewer 1 for providing a comprehensive and constructive 34-point review. Below, we address each of the 34 comments individually in a dedicated block.

Reviewer 1 — Comment R1.1 (Strengthen the Problem Formulation Section)

Reviewer's Suggestion (Restated): The Reviewer notes that the manuscript introduces the task at a high level without formally defining the problem in mathematical or system terms, and suggests adding a precise formulation with explicit variables, constraints, and assumptions to improve clarity and rigor.

Our Humble Response: We sincerely thank the Reviewer for this constructive critique. We humbly agree that our original submission described the control loop operationally without a formal mathematical optimization statement. In the revised **Section III-A**, we have explicitly stated the system variables (graph $G = (V, E)$, traffic demand matrix $W = [W_{st}]$, node service capacities $\mu = (\mu_i)_{i \in V}$, normalized stress state $S^{(k)} \in [0, 1]^N$, and node transit costs $C^{(k)} \in [1, 1 + \beta_I]^N$), formalized the underlying system assumptions (**Assumptions 1–4**), and introduced a rigorous constrained discrete-time bi-criteria optimization formulation in **Eq. (5)** balancing asymptotic peak stress $\limsup_{k \to \infty} \max_{i \in V} \widetilde{\kappa}_i(S^{(k)})$ and control-plane route-switching frequency $\bar{f}_{\text{sw}} \leq 1/T_d$.

Reviewer 1 — Comment R1.2 (Provide a Clearly Articulated Research Gap Statement)

Reviewer's Suggestion (Restated): The Reviewer points out that while the introduction lists related approaches, it does not explicitly isolate what remains unresolved in prior literature, and recommends adding a clear research gap statement.

Our Humble Response: We appreciate this helpful observation and humbly acknowledge that the research gap was previously implicit. In the revised **Section I** (and expanded **Section II-B**), we added a dedicated paragraph titled **Research Gap** that explicitly isolates the unresolved three-way tension in prior work: (i) reactive queue-based schemes suffer from queueing delay phase lag because they act only after congestion has already materialized; (ii) traffic-matrix-driven IGP weight optimization (e.g., Fortz & Thorup [26]) and memoryless centrality feedback trigger severe control-plane route flapping when applied online due to the discontinuous nature of shortest-path trees; and (iii) static centrality routing is open-loop and cannot adapt to dynamic traffic shifts. This directly motivates SRA's damped closed-loop topological control law.

Reviewer 1 — Comment R1.3 (Add a Comparative Summary Table Against Closest Prior Methods)

Reviewer's Suggestion (Restated): The Reviewer suggests adding a structured feature-by-feature comparison table against the closest prior methods to clarify whether the contribution of SRA is architectural, algorithmic, or parametric.

Our Humble Response: We thank the Reviewer for this great suggestion. We have expanded **Table I** in **Section I** into a comprehensive 6-column feature-by-feature comparison across five routing families: *Reactive Queue-Aware Routing*, *IGP Weight Tuning (Fortz–Thorup)*, *Static Centrality Routing*, *Data-Driven AI/DRL Routing*, and *Proposed SRA*. The table compares each method across *Control Signal*, *Feedback Loop Type*, *Adaptation Mode*, *Damping / Hysteresis Mechanism*, *Computational Complexity*, and *Primary Limitation*, clearly showing that SRA's novelty is both architectural and algorithmic (combining scale-invariant max-normalized structural stress, leaky-integrator state damping, and dwell-time hysteresis with provable set-valued stability).

Reviewer 1 — Comment R1.4 (Expand Theoretical Justification of the Core Algorithm)

Reviewer's Suggestion (Restated): The Reviewer remarks that several steps of the algorithm are presented operationally and recommends deeper analytical derivation or mathematical reasoning to reduce the perception of heuristic design.

Our Humble Response: We humbly accept this critique and thank the Reviewer for pushing us to strengthen our analytical foundation. We have substantially expanded **Sections III-C, III-D, and III-E** in the main text and provided complete, step-by-step formal proofs in **Supplementary Material Sections S-A through S-J**. Specifically, we rigorously justify: (i) why max-normalization confines the state trajectory to the invariant unit hypercube $X = [0, 1]^N$ across arbitrary network sizes (**Lemma 1, Section S-A**); (ii) why the Krasovskii-regularized inclusion $F(S) = (1 - \alpha)S + \alpha\mathcal{N}(S)$ is upper-semicontinuous with compact convex values and guarantees global attraction to a compact invariant set Ω (**Proposition 1 & Theorem 1, Sections S-B–S-C**); (iii) how the leaky integrator governs discrete-time Lyapunov tracking

energy dissipation and bounded switching jumps (**Theorem 2, Section S-D**); (iv) how minimum dwell time T_d enforces finite switching and bounds asymptotic chattering frequency by $1/T_d$ (**Proposition 2, Theorem 3, & Theorem 4, Sections S-F–S-H**); and (v) when structural peak shaving translates into guaranteed network throughput capacity expansion (**Lemma 2 & Proposition 3, Sections S-I–S-J**).

Reviewer 1 — Comment R1.5 (Include a Full Ablation Study Isolating Each Module)

Reviewer's Suggestion (Restated): The Reviewer notes that the manuscript does not clearly quantify the independent contribution of each component of SRA, and recommends a full ablation study to attribute the performance gains scientifically.

Our Humble Response: We completely agree with the Reviewer and apologize for omitting a modular ablation study in our initial submission. We have added a dedicated subsection (**Section IV-F**), quantitative summary table (**Table 4 / Table IV**), and multi-panel visualization (**Fig. 10**) that systematically isolate every constituent module of SRA across 30 independent trials on BA scale-free networks ($N = 100$, heavy load $\rho = 0.9$):

- **Removing the Leaky Integrator (*Memoryless Reactive BC*, $\alpha = 1, T_d = 1$)**: Causes violent route flapping ($\bar{f}_{\text{sw}} = 98.3\%$), amplifies peak stress by +31.04%, and degrades throughput to 53.39 ± 5.88 pkts/step (worse than static Dijkstra at 68.88 ± 5.57 pkts/step).
- **Removing Dwell-Time Gating (*SRA w/o Dwell*, $\alpha = 0.1, T_d = 1$)**: Restores high throughput (90.08 ± 0.38 pkts/step) via leaky-integrator smoothing, but still suffers from continuous control-plane route switching ($\bar{f}_{\text{sw}} = 98.3\%$).
- **Removing Max-Normalization (*Unnormalized SRA*, raw κ/N^2)**: Achieves 71.99% peak reduction and 90.05 ± 0.42 pkts/step on $N = 100$ BA graphs where initial $\kappa_{\max} \approx 2657$ happens to be on the order of $0.27N^2$, but loses scale invariance across different network sizes N and densities (Lemma 1).
- **Full SRA ($\alpha = 0.1, \beta_I = 10, T_d = 5$)**: Achieves 89.76 ± 0.76 pkts/step (+68.1% over $\alpha = 1$ and +30.3% over Dijkstra) while slashing route switching frequency by 79.7% (down to $\bar{f}_{\text{sw}} = 20.0\%$).

Reviewer 1 — Comment R1.6 (Introduce Statistical Validation Across Multiple Trials)

Reviewer's Suggestion (Restated): The Reviewer observes that reported results appear to be based on limited runs and recommends reporting mean, standard deviation/variance, and confidence intervals across multiple independent trials to improve credibility.

Our Humble Response: We thank the Reviewer for this essential standard of empirical rigor. In the revised manuscript, all new ablation, baseline comparison, heterogeneous/time-varying traffic, noise sensitivity, dwell-time, and scalability experiments in **Sections IV-F, IV-G, and IV-H** (**Tables 4 and 5 / Tables IV and V** and **Figs. 10, 11, and 13**) have been executed across **30 independent random seeds** per configuration. We explicitly report both the sample mean ± standard deviation (SD) and the normal-approximation **95% confidence intervals** ($\bar{x} \pm 1.96\, s/\sqrt{30}$) in **Table 4 (Table IV)** as well as shaded 95% CI bands and error bars across **Figs. 10, 11, and 13**.

Reviewer 1 — Comment R1.7 (Diversify Experimental Scenarios)

Reviewer's Suggestion (Restated): The Reviewer points out that validation under a single traffic/capacity configuration restricts generalizability and suggests adding diverse test scenarios to strengthen robustness claims.

Our Humble Response: We sincerely appreciate this suggestion. Beyond the stationary uniform-demand experiments on synthetic BA, WS, and ER graphs ($N = 100$), we have substantially diversified our experimental testbed in **Sections IV-G and IV-H** by adding:

1. **Skewed Heavy-Tailed Pareto Gravity Traffic Matrices** ($w_i \sim \text{Pareto}(1.5)$, $W_{st} \propto w_s w_t$) combined with **Heterogeneous Node Service Capacities** in **Section IV-G (Fig. 11(a))**;
2. **Non-Stationary Mid-Run Traffic Surge** in **Section IV-G (Fig. 11(b))**;
3. **Real-World Carrier & AS Backbone Topologies (`Internet AS`, $N = 60$; `Deltacom`, $N = 113$; and `UsCarrier`, $N = 158$)** as well as **Multi-Scale Synthetic Topologies ($N \in \{50, 100, 200, 300, 500\}$)** in **Section IV-H (Table 5 / Table V and Fig. 13)**.

Reviewer 1 — Comment R1.8 (Add Controlled Noise Injection Experiments)

Reviewer's Suggestion (Restated): The Reviewer notes that performance sensitivity to control-plane signal degradation was not explored, and suggests adding controlled noise injection experiments to validate practical reliability.

Our Humble Response: We thank the Reviewer for this insightful suggestion. In real-world software-defined or distributed telemetry environments, stress measurements may be corrupted by sampling lag or estimation noise. In **Section IV-G** and **Fig. 11(c)**, we added a controlled noise injection experiment in which multiplicative Gaussian telemetry noise $\widehat{\kappa}_i^{(k)} = \max\{0, \widetilde{\kappa}_i^{(k)}(1 + \xi_i^{(k)})\}$, $\xi_i^{(k)} \sim \mathcal{N}(0, \sigma^2)$, with $\sigma \in \{0\%, 5\%, 10\%, 20\%\}$ is injected at every control step. Because SRA's leaky integrator ($\alpha = 0.1$) acts as a first-order low-pass filter that attenuates zero-mean high-frequency variance by $\frac{\alpha}{2-\alpha} \approx 0.053$, Generalized Weighted SRA maintains virtually unchanged throughput (69.34 ± 12.14 pkts/step at $\sigma = 0\%$ vs. 69.64 ± 12.09 pkts/step at $\sigma = 20\%$) and packet delivery ratio (81.54% vs. 81.88%).

Reviewer 1 — Comment R1.9 (Clarify Dataset and Topology Representativeness)

Reviewer's Suggestion (Restated): The Reviewer remarks that the manuscript does not sufficiently justify whether the evaluated topologies and traffic demands reflect real-world network variability, which could limit external validity.

Our Humble Response: We humbly agree and have clarified the representativeness of both our topological benchmarks and traffic demand models in **Sections IV-A, IV-G, and IV-H**:

- **Topological Representativeness**: Synthetic Barabási–Albert (BA) scale-free graphs capture the heavy-tailed hub hierarchy of autonomous-system (AS) and data-center core networks; Watts–Strogatz (WS) small-world graphs capture regional mesh networks with local clustering and long-range shortcuts; and Erdős–Rényi (ER) graphs serve as a homogeneous random baseline. To directly ground external validity in operational wide-area carrier networks, we additionally evaluated three production-scale carrier/AS topologies in

Section IV-H (Table 5 / Table V): **Internet AS** ($N = 60, E = 85$, +18.91% throughput gain), **Deltacom** ($N = 113, E = 254$, +52.24% throughput gain), and **UsCarrier** ($N = 158, E = 536$, +81.07% throughput gain).

- **Traffic Representativeness**: In **Section IV-G**, we complemented uniform all-pairs traffic with a heavy-tailed **Pareto Gravity Traffic Matrix** ($w_i \sim \text{Pareto}(1.5)$, $W_{st} \propto w_s w_t$), which reflects the well-known skewed hotspot demand distribution of real ISP and data-center workloads.

Reviewer 1 — Comment R1.10 (Provide Computational Complexity Analysis)

Reviewer's Suggestion (Restated): The Reviewer points out that without formal time and memory complexity analysis, scalability claims remain unsubstantiated, and recommends adding a clear complexity breakdown.

Our Humble Response: We thank the Reviewer for highlighting this omission. In the revised **Section III-B (Algorithm 1)** and **Section IV-H**, we added an explicit time and space complexity analysis:

- **Time Complexity**: At active routing recomputation epochs ($k \bmod T_d = 0$), computing all-pairs shortest paths and node centrality via Brandes' algorithm on a positively weighted graph requires $O(N(E + N \log N))$ time, followed by $O(N)$ state and cost updates. During the $T_d - 1$ frozen cooling steps within each dwell block, shortest-path trees and stress vectors are cached and reused, requiring only $O(N)$ vector arithmetic per step. Hence, the amortized per-step control-plane time complexity is $O\left(\frac{N(E+N\log N)}{T_d} + N\right)$.
- **Space Complexity**: Working memory for Brandes' traversal and state vectors is $O(N + E)$ ($O(N)$ for node states/costs and $O(E)$ for adjacency lists), and storing all-pairs next-hop routing tables takes $O(N^2)$ space.

Reviewer 1 — Comment R1.11 (Report Runtime Benchmarks on Specified Hardware)

Reviewer's Suggestion (Restated): The Reviewer recommends reporting concrete execution time benchmarks on specified hardware to ground the method's implementation feasibility in practical constraints.

Our Humble Response: We gladly followed this recommendation. In **Section IV-H**, **Table 5 (Table V)**, and **Fig. 13(b)**, we report the single-thread wall-clock execution time per Brandes routing/centrality evaluation measured on an **Intel Xeon Gold 6248R CPU @ 3.00 GHz** server (Python 3.10):

- **Real-World Carrier/AS Backbones (`Internet AS` $N = 60$, `Deltacom` $N = 113$, `UsCarrier` $N = 158$)**: 8.52 ± 0.68 ms, 36.96 ± 0.95 ms, and 77.33 ± 3.30 ms per evaluation (1.70 ms, 7.39 ms, and 15.47 ms amortized per step at $T_d = 5$);
- **Synthetic BA/WS ($N = 100$)**: 32.12–32.45 ms per evaluation (≈ 6.4 ms amortized per step at $T_d = 5$);
- **Large Scale-Free BA ($N = 200, 300, 500$)**: 140.60 ms, 320.68 ms, and 902.95 ms per evaluation ($\approx$ 180.6 ms amortized per step at $N = 500, T_d = 5$), which is well within standard SDN sub-second to multi-second control-plane telemetry intervals.

Reviewer 1 — Comment R1.12 (Improve Parameter Selection Justification)

Reviewer's Suggestion (Restated): The Reviewer notes that hyperparameters (α, β_I, T_d) appear tuned empirically and requests a principled control-theoretic rationale and sensitivity analysis for their selection.

Our Humble Response: We thank the Reviewer for this constructive comment and humbly agree that the physical meaning of (α, β_I, T_d) needed clearer justification. In **Section III-D**, **Section IV-B**, and **Section IV-H**, we added a principled selection guideline grounded in discrete-time control theory:

1. **Relaxation Rate** $\alpha \in [0.08, 0.25]$ **(Default** $\alpha = 0.1$**)**: Governs the exponential memory horizon $\tau_{\text{relax}} = -1/\ln(1-\alpha) \approx 1/\alpha \approx 10$ epochs and bounds asymptotic tracking energy $\limsup_{k\to\infty} E_k \leq \bar{\Delta}^2/\alpha^2$ (**Theorem 2**) while attenuating telemetry noise variance by $\alpha/(2-\alpha)$.
2. **Sensitivity Gain** $\beta_I \in [8, 20]$ **(Default** $\beta_I = 10$**)**: Scales the maximum node transit cost $1 + \beta_I$ relative to the baseline hop count ($\bar{L} \approx 2.4$–3.1 hops in $N = 100$ graphs) so that an overloaded hub with $S_u \to 1$ induces a detour penalty comparable to 2–3 alternate hops without causing excessive path stretch.
3. **Minimum Dwell Time** $T_d \in [3, 8]$ **(Default** $T_d = 5$**)**: Selected as approximately $0.5\,\tau_{\text{relax}}$ so that within each dwell block, the tracking error $\|S^{(k)} - q^{(k)}\|_2$ contracts by $(1-\alpha)^{T_d} \approx 0.59$ (absorbing $> 40\%$ of the transient gap) while capping route switching frequency at $1/T_d = 20\%$.

Reviewer 1 — Comment R1.13 (Include Hyperparameter Sensitivity Plots)

Reviewer's Suggestion (Restated): The Reviewer suggests including hyperparameter sensitivity plots across parameter ranges to demonstrate stability and reduce the impression of overfitting.

Our Humble Response: We thank the Reviewer for this suggestion. We have included two dedicated sensitivity visualizations in the revised manuscript:

1. **Joint 3D Hyperparameter Sensitivity Surfaces over** (α, β_I) **(Fig. 12 in Section IV-H)**: Displays peak centrality reduction $\Delta\kappa_{\max}$, average path stretch $\Delta\bar{L}$, and convergence iterations over the grid $\alpha \in [0.05, 0.50] \times \beta_I \in [2.0, 20.0]$, revealing a broad, flat performance plateau across $\alpha \in [0.08, 0.25]$ and $\beta_I \in [8.0, 20.0]$ where peak stress reduction consistently exceeds 80%.
2. **Dwell-Time Sensitivity Curve over** $T_d \in \{1, 2, 3, 5, 8, 10\}$ **(Fig. 13(a) in Section IV-H)**: Plots network throughput, packet delivery ratio, and route switching frequency across T_d (with 95% CIs over 30 seeds), showing stable high throughput across $T_d \in [1, 8]$ (89.19–90.07 pkts/step, PDR 99.07%–99.99%) and hyperbolic reduction in route switching $\bar{f}_{\text{sw}} \approx 1/T_d$.

Reviewer 1 — Comment R1.14 (Clarify Boundary Conditions and Assumptions)

Reviewer's Suggestion (Restated): The Reviewer notes that several model assumptions and boundary conditions were left implicit and recommends stating them explicitly to improve transparency.

Our Humble Response: We humbly agree and have stated all model assumptions and boundary conditions explicitly in **Section III-A (Assumptions 1–4)** and **Section III-C (Lemma 1)**. Specifically, we clarify: (i) the topological requirement of alternate vertex-disjoint paths around bottlenecks (**Assumption 1**); (ii) time-scale separation between control epochs and packet queuing (**Assumption 2**); (iii) deterministic tie-breaking or ECMP ensuring a finite set of $M < \infty$ shortest-path configurations (**Assumption 3**); (iv) positive node capacities $\mu_i > 0$ (**Assumption 4**); and (v) the exact initial state boundary condition $S^{(0)} \in [0, 1]^N$ (or perturbed initialization $S^{(0)} \in [-1/\beta_I + \epsilon, \infty)^N$ ensuring strictly positive node costs $C_i^{(0)} \geq 1 + \beta_I \epsilon > 0$ in **Lemma 1**).

Reviewer 1 — Comment R1.15 (Expand Discussion of Failure Cases)

Reviewer's Suggestion (Restated): The Reviewer points out that the manuscript focuses on positive results and omits situations where SRA underperforms or provides limited benefit, recommending an explicit discussion of failure cases.

Our Humble Response: We sincerely thank the Reviewer for this important reminder of scientific balance. In the revised manuscript, we created a dedicated subsection (**Section V-B: *Explicit Assumptions, Failure Cases, and Limitations***) that candidly analyzes three concrete scenarios where SRA provides negligible gain or underperforms:

1. **Homogeneous Regular Topologies (e.g., 2D Tori / Regular Lattices)**: Because every node already shares an identical baseline centrality ($\kappa_{\max} \approx \bar{\kappa}$), there is no structural stress concentration to relax; SRA yields $< 2\%$ throughput change while incurring slight path-stretch overhead at low loads.
2. **Ultra-Sparse Cut-Vertex / Single-Bridge Topologies (Trees / Dumbbell Graphs)**: When a bottleneck node v_{cut} is a topological cut-vertex disconnecting the graph, every (s, t) path across the cut must traverse v_{cut} regardless of its node cost $C_{v_{\text{cut}}}$. Consequently, the load-shedding term in Lemma 2 is identically zero ($\Delta^-_{v_{\text{cut}}} = 0$), and no routing algorithm can bypass the physical cut.
3. **Overly Aggressive Penalty $\beta_I \gg 20$ or Excessive Dwell Time $T_d \geq 10$**: As shown in **Fig. 12** and **Fig. 13(a)**, setting β_I too high inflates path stretch $\Delta\bar{L}$, while setting $T_d = 10$ slows transient adaptation during initial congestion buildup, reducing BA throughput from 89.76 ($T_d = 5$) to 77.87 pkts/step ($T_d = 10$).

Reviewer 1 — Comment R1.16 (Include Visual Comparison Under Challenging Scenarios)

Reviewer's Suggestion (Restated): The Reviewer suggests adding comparative visual figures under difficult operational conditions (such as non-stationary surges, skewed demands, and noisy telemetry) to provide a more balanced evaluation.

Our Humble Response: Following the Reviewer's suggestion, we added the 3-panel comparative visualization in **Fig. 11 (Section IV-G)** evaluating *Static Dijkstra*, *Topology-Only SRA*, and *Generalized Weighted SRA* under simultaneous challenging conditions: (a) skewed Pareto gravity demand and heterogeneous node capacities across offered loads; (b) a sudden $2\times$ non-stationary traffic surge showing real-time throughput tracking and post-surge recovery; and (c) degraded control-plane telemetry with multiplicative Gaussian noise up to $\sigma = 20\%$.

Reviewer 1 — Comment R1.17 (Improve Figure Readability and Annotation Quality)

Reviewer's Suggestion (Restated): The Reviewer notes that some figures in the original manuscript lacked sufficient labeling clarity or had missing visual panels, and recommends improving figure readability and annotations.

Our Humble Response: We humbly apologize for the figure omissions and annotation issues in the original submission. We have restored the three omitted vector figures—**Fig. 6** (`ren_fig2_lyapunov.pdf`), **Fig. 7** (`ren_fig3_dwell_time.pdf`), and **Fig. 12** (`sensitivity_3d_surfaces_full.pdf`)—removed the stray text artifact '[cite: 276, 277]' from **Fig. 8**, and generated high-resolution vector PDFs for all new figures (**Figs. 10, 11, and 13**) with clear axis units, shaded 95% confidence bands, annotated surge windows, and self-contained captions.

Reviewer 1 — Comment R1.18 (Reorganize Methodology for Logical Flow)

Reviewer's Suggestion (Restated): The Reviewer remarks that certain parts of the methodology interleaved operational explanation and results, and recommends reorganizing Section III for a cleaner logical progression.

Our Humble Response: We thank the Reviewer for this structural advice. We have reorganized **Section III** into a strict five-stage logical progression:

1. **Section III-A**: Network Model, Explicit Assumptions 1–4, Generalized Demand-/Capacity-Weighted Stress $\widetilde{\kappa}_i(S; W, \mu)$, and Formal Bi-Criteria Optimization Problem (Eqs. (1)–(5));
2. **Section III-B**: Closed-Loop SRA Control Law and Stepwise Pseudocode (**Algorithm 1**) with complexity analysis (Eqs. (6)–(7));
3. **Section III-C**: Krasovskii Set-Valued Regularization, Box Invariance (**Lemma 1**), Global Attraction to Compact Limit Set Ω (**Proposition 1, Theorem 1, Remark 2**), and Discrete-Time Drop-Jump Energy Bound (**Theorem 2, Corollary 1**) (Eqs. (8)–(14));
4. **Section III-D**: Minimum Dwell-Time Block Gating, Finite-Horizon Switching Bound (**Proposition 2**), Blockwise Contraction (**Theorem 3**), and Chattering Suppression (**Theorem 4**) (Eqs. (15)–(17));
5. **Section III-E**: Theoretical Bridge from Structural Peak Shaving (**Lemma 2**) to Network Capacity Expansion (**Proposition 3**) (Eqs. (18)–(23)).

Reviewer 1 — Comment R1.19 (Reduce Repetitive Narrative in Discussion)

Reviewer's Suggestion (Restated): The Reviewer points out that portions of the discussion section reiterated earlier explanations without adding new analytical depth, and suggests condensing repetitive text.

Our Humble Response: We humbly agree and have substantially condensed **Section V-A** (as well as repetitive introductory paragraphs in Sections I–IV), removing redundant restatements of the algorithm workflow and focusing the discussion sharply on the control-theoretic synthesis of spatial relaxation and temporal dwell-time regularization. Together with moving the full mathematical proofs to the standalone **Supplementary Material (`sm.pdf`)**, this reduced the

China Communications manuscript from 31 pages to 22 pages (and the IEEEtran version from 25 pages to 19 pages).

Reviewer 1 — Comment R1.20 (Condense Conclusion to Focus on Quantified Findings)

Reviewer's Suggestion (Restated): The Reviewer recommends condensing the conclusion so that it synthesizes concrete quantified findings rather than restating the methodological workflow.

Our Humble Response: We thank the Reviewer for this suggestion. We have rewritten **Section V-C (Conclusion)** to concisely synthesize the key quantified findings of our study: (i) an 83.76% reduction in peak betweenness centrality on BA scale-free networks ($N = 100$), driving a +31.87% throughput gain at $N = 100$ (+30.3% in the 30-seed ablation campaign at $\rho = 0.9$) and +108.33% seedwise relative throughput gain at $N = 500$; (ii) a 79.7% reduction in control-plane route switching frequency (20.0% vs. 98.3%) via dwell-time gating $T_d = 5$; and (iii) a +20.28% throughput gain and 55.6% generalized peak stress reduction under skewed Pareto gravity traffic and heterogeneous capacities.

Reviewer 1 — Comment R1.21 (Provide Clearer Algorithm Pseudocode)

Reviewer's Suggestion (Restated): The Reviewer suggests providing a clearer, formal stepwise algorithm block to enhance reproducibility.

Our Humble Response: We have thoroughly revised **Algorithm 1** in **Section III-B** to explicitly detail every step of the closed-loop execution over $k = 0, \ldots, K_{\max} - 1$: (1) checking the dwell-time gate ($k \bmod T_d == 0$) to either recompute shortest paths $\mathcal{P}^{(k)}$ and topological/generalized stress via Brandes' algorithm or reuse cached paths/stresses during cooling steps; (2) handling the $\sigma_{st} = 0$ guard; (3) computing the max-normalized stress feedback $q^{(k)} = \text{norm}(\kappa^{(k)}) \in [0, 1]^N$; (4) updating the leaky-integrator state $S^{(k+1)} = (1 - \alpha)S^{(k)} + \alpha q^{(k)}$; and (5) updating the node transit costs $C^{(k+1)} = \mathbf{1} + \beta_I S^{(k+1)}$, accompanied by explicit per-step time and memory complexity annotations.

Reviewer 1 — Comment R1.22 (Explicitly State Limitations in a Dedicated Subsection)

Reviewer's Suggestion (Restated): The Reviewer notes that limitations were previously scattered and understated, and recommends consolidating them into a dedicated subsection to improve scientific honesty.

Our Humble Response: We wholeheartedly agree and have created a dedicated subsection, **Section V-B (*Explicit Assumptions, Failure Cases, and Limitations*)**, which consolidates all model assumptions, topological failure modes (homogeneous lattices/tori and single-bridge cut-vertex networks), telemetry lag constraints, and centralized Brandes computation scaling limits ($O(N(E + N \log N))$), along with concrete directions for distributed/approximate centrality extensions.

Reviewer 1 — Comment R1.23 (Compare Against Stronger Baselines)

Reviewer's Suggestion (Restated): The Reviewer observes that the original comparisons were limited to Dijkstra and LIRP-D, and recommends benchmarking against stronger, more advanced routing baselines.

Our Humble Response: We thank the Reviewer for this important suggestion. In **Section IV-F (Table 4 / Table IV and Fig. 10)**, we added two closely related traffic-engineering and centrality-routing baselines alongside *Dijkstra* and *LIRP-D*:

1. **Online IGP Link-Weight Tuning (Fortz–Thorup [26])**: Dynamically adjusts routing weights using convex piecewise-linear utilization penalties. While it achieves high throughput (90.08 ± 0.38 pkts/step), it suffers from continuous route flapping ($\bar{f}_{\mathrm{sw}} = 98.3\%$).
2. **Static Betweenness Centrality Routing [19]**: Assigns fixed node costs proportional to initial topological centrality ($C = \mathbf{1} + \beta_I \mathrm{norm}(\kappa^{(0)})$). While effective under stationary uniform traffic (89.96 ± 0.51 pkts/step), it is strictly open-loop and cannot adapt to secondary bottlenecks or non-stationary traffic surges.

Reviewer 1 — Comment R1.24 (Explain Metric Selection Rationale)

Reviewer's Suggestion (Restated): The Reviewer points out that the manuscript did not explicitly justify why specific evaluation metrics were chosen, and recommends explaining the metric selection rationale.

Our Humble Response: We thank the Reviewer for pointing this out. In **Sections IV-D and IV-F**, we added a clear rationale explaining how our six evaluation metric categories jointly capture both data-plane efficiency and control-plane stability: (1) **Effective Throughput** and (2) **Packet Delivery Ratio (PDR) / Drop Rate** measure data-plane capacity and congestion loss; (3) **Mean and P95 Tail Latency** capture both typical path-stretch cost and worst-case queuing delay; (4) **Peak Stress Reduction ($\Delta\kappa_{\max}$ / $\max_i \widetilde{\kappa}_i$) and BC Standard Deviation** quantify structural load balancing across nodes; (5) **Route Switching Frequency ($\bar{f}_{\mathrm{sw}}$)** measures control-plane routing table stability and chattering suppression; and (6) **Wall-Clock Runtime (ms/eval)** quantifies computational overhead.

Reviewer 1 — Comment R1.25 (Include Additional Evaluation Metrics)

Reviewer's Suggestion (Restated): The Reviewer recommends incorporating additional complementary evaluation metrics beyond average throughput and average latency to provide a more comprehensive assessment.

Our Humble Response: Following the Reviewer's recommendation, we added five complementary metrics in **Tables 4 and 5 (Tables IV and V)** and **Figs. 10, 11, and 13**: (i) **95th-Percentile (P95) Tail Packet Latency**; (ii) **Packet Delivery Ratio (PDR %)** and **Steady-State Drop Rate (%)**; (iii) **Control-Plane Route Switching Frequency ($\bar{f}_{\mathrm{sw}}$)**; (iv) **Peak Generalized Demand-/Capacity-Normalized Stress ($\max_i \widetilde{\kappa}_i$)**; and (v) **Per-Evaluation Wall-Clock Runtime (ms/eval)**.

Reviewer 1 — Comment R1.26 (Discuss Scalability to Larger or More Complex Inputs)

Reviewer's Suggestion (Restated): The Reviewer asks for an evaluation and discussion of how SRA scales to larger network sizes and more complex real-world topologies.

Our Humble Response: We have added a comprehensive scalability study in **Section IV-H (Table 5 / Table V and Fig. 13(b))** spanning synthetic topologies of size $N \in \{50, 100, 200, 300, 500\}$ as well as three realistic carrier/AS backbones: **Internet AS** ($N = 60, E = 85$, +18.91% throughput gain), **Deltacom** ($N = 113, E = 254$, +52.24% throughput gain), and **UsCarrier** ($N = 158, E = 536$, +81.07% throughput gain). Remarkably, as scale-free networks grow from $N = 50$ to $N = 500$, central hub concentration under static shortest-path routing becomes increasingly severe, causing SRA's seedwise relative throughput advantage over Dijkstra to grow monotonically from +10.41% ($N = 50$) and +31.87% ($N = 100$) to +60.68% ($N = 200$), +79.65% ($N = 300$), and +108.33% ($N = 500$).

Reviewer 1 — Comment R1.27 (Clarify Evaluation Protocol and Simulation Details)

Reviewer's Suggestion (Restated): The Reviewer recommends clarifying all simulation protocol details (such as horizon lengths, warm-up windows, queue capacities, and packet generation rules) to ensure full transparency.

Our Humble Response: We have documented all simulation protocol details in **Section IV-A**, **Table 2 (Table II)**, and the captions of **Tables 4–5 (Tables IV–V)**. Specifically, we clarify the discrete-event packet simulator settings: baseline normalized service rate $\mu = 1.0$ (with packet processing capacity 10 packets/timestep and FIFO buffer capacity $B_{\max} = 50$ packets), warmup and measurement windows, and 30 independent random seeds per configuration across all ablation, noise, sensitivity, and scalability benchmarks.

Reviewer 1 — Comment R1.28 (Improve Mathematical Notation Consistency)

Reviewer's Suggestion (Restated): The Reviewer notes minor inconsistencies in mathematical notation across sections and appendices and recommends unifying all symbols.

Our Humble Response: We thank the Reviewer for catching this and humbly apologize for the inconsistent notation in the original draft. We have audited and unified all symbols across the main text (`main_ccjnl.tex` / `main.tex`) and the Supplementary Material (`sm.tex`): $k \in \mathbb{N}$ denotes the discrete control epoch, $E = |E|$ denotes edge count, M denotes the number of distinct shortest-path configurations, $S_i^{(k)} \in [0, 1]$ denotes the normalized structural stress state, $q_i^{(k)} = \mathrm{norm}(\kappa_i^{(k)}) \in [0, 1]$ denotes the normalized instantaneous stress feedback, $E_k = \|S^{(k)} - q^{(k)}\|_2^2$ denotes the discrete-time tracking energy, μ_i denotes node service rate, $B_{\max}$ denotes node buffer capacity, and T_d denotes the minimum dwell time.

Reviewer 1 — Comment R1.29 (Strengthen Connection Between Theory and Experiments)

Reviewer's Suggestion (Restated): The Reviewer suggests explicitly linking each theoretical theorem/proposition to the corresponding empirical figures and tables in Section IV.

Our Humble Response: We appreciate this suggestion and have added explicit cross-references throughout **Sections III and IV** connecting each theoretical result to its empirical verification:

- **Lemma 1 & Theorem 1** (Box invariance and attraction to compact limit set Ω) $\leftrightarrow$ verified by the macroscopic convergence curves in **Fig. 6(a)** and phase-space trajectories in **Fig. 7(a)**;
- **Theorem 2** (Discrete-time Drop-Jump tracking energy bound $E_{k+1} \leq (1-\alpha)E_k + \bar{\Delta}^2/\alpha$) $\leftrightarrow$ directly visualized in **Fig. 6** (`ren_fig2_lyapunov.pdf`);
- **Proposition 2, Theorem 3, & Theorem 4** (Dwell-time chattering suppression $\bar{f}_{\text{sw}} \leq 1/T_d$) $\leftrightarrow$ verified by **Fig. 7(b)**, **Table 4 (Table IV)**, and **Fig. 13(a)**;
- **Lemma 2 & Proposition 3** (Peak stress reduction and inverse-peak capacity scaling) $\leftrightarrow$ verified by the linear regression $R^2 = 0.951$ in **Fig. 5**, the throughput gains in **Table 4 (Table IV) / Fig. 10**, and the generalized stress experiments in **Fig. 11**.

Reviewer 1 — Comment R1.30 (Provide Ablation on Preprocessing or Normalization Steps)

Reviewer's Suggestion (Restated): The Reviewer recommends evaluating the impact of the max-normalization preprocessing step ($\text{norm}(\cdot)$*) via an ablation comparison against unnormalized feedback.*

Our Humble Response: We have included this exact ablation in **Section IV-F** and **Table 4 (Table IV)** by comparing **Full SRA** (with scale-invariant max-normalization $q^{(k)} = \kappa^{(k)}/\kappa^{(k)}_{\max} \in [0,1]^N$) against **Unnormalized SRA** (which feeds raw $\kappa^{(k)}/N^2$ into the state update). We show both theoretically (**Lemma 1**) and empirically that while $\kappa^{(k)}/N^2$ happens to have an $O(1)$ peak on $N = 100$ BA hubs ($\kappa_{\max} \approx 2657$, achieving 71.99% peak reduction and 90.05 ± 0.42 pkts/step), its effective loop gain $\beta_I \kappa_{\max}/N^2$ varies by orders of magnitude across different network sizes N and densities, whereas max-normalization guarantees a parameter-invariant state space $X = [0,1]^N$ across all network sizes and topologies.

Reviewer 1 — Comment R1.31 (Address Potential Evaluation Bias or Hyperparameter Overfitting)

Reviewer's Suggestion (Restated): The Reviewer asks the authors to ensure that experimental gains are not an artifact of per-scenario hyperparameter tuning or evaluation bias.

Our Humble Response: We thank the Reviewer for raising this crucial methodological point. In **Sections IV-A, IV-F, IV-G, and IV-H**, we explicitly confirm that **one single fixed default hyperparameter tuple** ($\alpha = 0.1, \beta_I = 10, T_d = 5$) is used unchanged across all topologies (BA, WS, ER, `Internet AS`, `Deltacom`, `UsCarrier`), all network sizes ($N \in [50, 500]$), and all traffic/noise regimes without any per-instance retuning, and every benchmark is evaluated over 30 independent random seeds generated fresh for testing.

Reviewer 1 — Comment R1.32 (Improve Figure Captions to Be Self-Contained)

Reviewer's Suggestion (Restated): The Reviewer suggests expanding the figure and table captions so that readers can understand the experimental setup, metrics, and key takeaways directly from the captions.

Our Humble Response: We have thoroughly expanded the captions of all figures (**Figs. 1–13**) and tables (**Tables 1–5 / Tables I–V**) in the revised manuscript so that each caption explicitly states the network topology, size N, offered load ρ, number of independent trials (30 seeds), confidence interval definition, and principal quantitative takeaway.

Reviewer 1 — Comment R1.33 (Update Bibliography with Relevant Classical and Recent Literature)

Reviewer's Suggestion (Restated): The Reviewer recommends updating the bibliography to include foundational traffic engineering, centrality computation, and set-valued control references alongside recent routing literature.

Our Humble Response: We have updated the bibliography to 40 references, incorporating foundational works on IGP link-weight optimization (Fortz & Thorup, *IEEE JSAC* [26]), fast betweenness centrality computation (Brandes, *J. Math. Sociol.* [27]), and differential inclusions / upper-semicontinuous set-valued maps (Aubin & Cellina [28]), alongside recent works in *IEEE JSAC*, *IEEE TCOM*, *IEEE TNSE*, and *IEEE/ACM ToN*.

Reviewer 1 — Comment R1.34 (Provide Code and Implementation Details for Reproducibility)

Reviewer's Suggestion (Restated): The Reviewer requests that source code and implementation details be made available to facilitate independent reproduction of all results.

Our Humble Response: In full support of open science and reproducibility, we have publicly released the complete simulation framework, baseline implementations, 30-seed campaign runner scripts, raw JSON/CSV result logs, plotting scripts, and LaTeX manuscript/supplementary sources in our GitHub repository: https://github.com/unicorners-compin/SRA-Resilient-Network-Governance (cited in **Section V-C**).

Part III: Individual Responses to Reviewer 2 (Comments R2.1 – R2.5)

We sincerely thank Reviewer 2 for an extraordinarily thorough, rigorous, and constructive review. Your sharp mathematical audit of Appendices A, C, D, I, and J and your identification of the omitted figure references in Section IV have helped us eliminate several critical flaws. Below, we respond to each of your comments and sub-points individually.

Reviewer 2 — Comment R2.1 (Novelty Beyond EWMA + Dwell-Time Filtering and Comparison Against Centrality/IGP Baselines)

Reviewer's Suggestion (Restated): The Reviewer asks the authors to articulate what is fundamentally new in SRA beyond applying an EWMA filter (α) and a dwell-time threshold (T_d) to normalized betweenness centrality, and recommends comparing SRA against a static centrality-weighted routing scheme and a dynamic weight-tuning/congestion-smoothing routing baseline to isolate where the performance gains originate.

Our Humble Response: We sincerely thank Reviewer 2 for this penetrating question. We humbly agree that our original manuscript did not sufficiently isolate the closed-loop dynamical

contribution of SRA from a simple cascade of standard filters, nor did it benchmark against static centrality or online IGP weight tuning. We have addressed both aspects in detail:

1. **Conceptual and Theoretical Novelty Beyond Simple Filtering (Section I, Table I, Section II-B, and Section III-C/D)**: When shortest-path routing $\mathcal{R}(G, C)$ is placed in a closed feedback loop with topological centrality $\kappa(S)$, the resulting map is *piecewise constant and discontinuous* across polyhedral tie-breaking boundaries. Consequently:

   - **Memoryless Reactive Centrality ($\alpha = 1$)** or **Online IGP Weight Tuning (Fortz–Thorup [26])** directly actuates discontinuous path switches at every step, inducing severe boundary chattering (98.3% route switching rate);
   - **Open-Loop Static Centrality Routing** computes node centrality only once on the unweighted graph ($k = 0$), making it blind to secondary bottlenecks created after traffic shifts and unable to adapt to non-stationary demand surges;
   - **SRA's Fundamental Contribution** lies in the *coupled hybrid control architecture* and its *non-smooth set-valued stability theory*: we prove that embedding max-normalized structural stress into the leaky-integrator inclusion $S^{(k+1)} \in (1-\alpha)S^{(k)} + \alpha\mathcal{N}(S^{(k)})$ guarantees a compact global attracting limit set $\Omega \subset [0,1]^N$ (**Theorem 1**) and an exact discrete-time Drop-Jump tracking energy bound $E_{k+1} \leq (1-\alpha)E_k + \bar{\Delta}^2/\alpha$ (**Theorem 2**), while pairing α with a dwell-time gate $T_d \approx 0.5/\alpha$ suppresses boundary limit-cycle chattering on Ω (**Theorems 3–4**) and provably expands network capacity (**Lemma 2, Proposition 3**).

2. **Empirical Isolation Against Static BC, Memoryless Reactive BC, and Online IGP Weight Tuning (Section IV-F, Table 4 / Table IV, and Fig. 10)**: Across 30 independent trials on BA scale-free networks ($N = 100, \rho = 0.9$):

   - **Memoryless Reactive BC ($\alpha = 1, T_d = 1$)** suffers from 98.3% route flapping, amplifies peak centrality by +31.04%, and collapses throughput to 53.39 ± 5.88 pkts/step (below static Dijkstra's 68.88 ± 5.57 pkts/step).
   - **Static BC (One-Shot Open-Loop)** achieves 89.96 ± 0.51 pkts/step under stationary uniform traffic, but cannot adapt when traffic matrices or capacities are skewed/time-varying (**Section IV-G**).
   - **Online IGP Weight Tuning (Fortz–Thorup)** reaches 90.08 ± 0.38 pkts/step, but switches routing tables at 98.3% **of control epochs**.
   - **Full SRA ($\alpha = 0.1, \beta_I = 10, T_d = 5$)** achieves 89.76 ± 0.76 pkts/step (99.65% PDR) while slashing control-plane route switching by 79.7% (20.0% vs. 98.3%).

Reviewer 2 — Comment R2.2(a) (Mathematical Rigor in Lemma 1 / Supplement Section S-A and Theorem 1 / Supplement Section S-C: Undefined $\delta^{\pm}_{k,i}$ and Attraction to Compact Limit Set Ω vs. Unique Equilibrium)

Reviewer's Suggestion (Restated): The Reviewer points out that in Appendix A, $\delta^{+}_{k,i}$ and $\delta^{-}_{k,i}$ appear abruptly without definition, and that in Theorem 1 and Appendix C, the proof establishes attraction to a compact invariant limit set Ω rather than convergence to a unique fixed-point equilibrium.

Our Humble Response: We are deeply grateful to Reviewer 2 for catching both of these flaws. We humbly admit that the definitions of $\delta^{\pm}_{k,i}$ were accidentally truncated during an earlier editing pass, and that titling Theorem 1 "Universal Convergence to Equilibrium" was mathematically

imprecise for a system with multi-path switching boundaries. We have completely repaired both items in **Section III-C** and **Supplementary Material Sections S-A, S-B, and S-C**:

1. **Explicit Definition of Box-Violation Sequences in Lemma 1 (Supplement Section S-A)**: We explicitly defined the coordinate-wise upper and lower box-violation excesses $\delta_{k,i}^{+} := \max\{S_i^{(k)} - 1,\, 0\}$ and $\delta_{k,i}^{-} := \max\{-S_i^{(k)},\, 0\}$. Since $q_i^{(k)} \in [0,1]$, any state initialized in $X = [0,1]^N$ satisfies $\delta_{k,i}^{\pm} \equiv 0$ for all $k \geq 0$ (positive invariance), and any perturbed initial state $S^{(0)} \in [-1/\beta_I + \epsilon, \infty)^N$ satisfies $\delta_{k+1,i}^{\pm} \leq (1-\alpha)\delta_{k,i}^{\pm} = (1-\alpha)^{k+1}\delta_{0,i}^{\pm} \to 0$.
2. **Rigorous Nested-Compact-Set Construction of the Global Attractor Ω in Theorem 1 & Remark 2 (Section III-C and Supplement Sections S-B–S-C)**: We retitled **Theorem 1** as *"Global Convergence to a Compact Invariant Attracting Set"* and constructed the global attractor on the compact state space $X = [0,1]^N$ via the nested intersection $\Omega := \bigcap_{n=0}^{\infty} F^n(X) \subset X$, where $F(S) = (1-\alpha)S + \alpha\mathcal{N}(S)$ is upper-semicontinuous with non-empty compact convex values (**Proposition 1**). By Aubin–Cellina [28], Ω is non-empty, compact, and strictly invariant ($F(\Omega) = \Omega$), and every trajectory satisfies $\lim_{k\to\infty} \operatorname{dist}(S^{(k)}, \Omega) = 0$. Furthermore, we added **Remark 2** to explicitly distinguish global attraction to the compact set Ω (which may contain bounded tie-breaking limit cycles near switching boundaries) from convergence to a unique point equilibrium $S^\star = c$ (which holds conditionally within a single stable routing stratum in **Corollary 1**).

Reviewer 2 — Comment R2.2(b) (Mathematical Rigor in Theorem 2 / Supplement Section S-D: Discontinuous $\kappa(S)$, Normalized Feedback $q^{(k)}$, and Exact Discrete-Time Energy Inequality)

Reviewer's Suggestion (Restated): The Reviewer notes that Theorem 2 and Appendix D previously mixed unnormalized $\kappa(S)$ with normalized $\operatorname{norm}(\kappa(S))$ *and relied on a heuristic continuous-time analogy ($\dot{V}$) and an implicit projection choice c_k, requesting a rigorous discrete-time derivation aligned with the implemented algorithm.*

Our Humble Response: We sincerely thank Reviewer 2 for this precise mathematical critique. We humbly acknowledge that mixing $\kappa(S)$ and $\operatorname{norm}(\kappa(S))$ and invoking a continuous-time analogy weakened the rigor of Theorem 2. In the revised **Section III-C (Theorem 2)** and **Supplementary Material Section S-D**, we have completely rewritten the statement and proof entirely in discrete time using the exact implemented normalized feedback $q^{(k)} := \operatorname{norm}(\kappa^{(k)}) \in [0,1]^N$ and the discrete-time tracking energy $E_k := \|S^{(k)} - q^{(k)}\|_2^2$:

- From the exact algebraic identity $S^{(k+1)} - q^{(k+1)} = (1-\alpha)(S^{(k)} - q^{(k)}) + (q^{(k)} - q^{(k+1)})$, expanding the squared Euclidean norm and applying Weighted Young's inequality $2\langle a, b\rangle \leq \lambda\|a\|_2^2 + \lambda^{-1}\|b\|_2^2$ with $\lambda = \frac{\alpha}{1-\alpha} > 0$ yields the exact one-step Drop-Jump inequality:

$$E_{k+1} \leq (1-\alpha)E_k + \frac{1}{\alpha}\left\|q^{(k)} - q^{(k+1)}\right\|_2^2 \leq (1-\alpha)E_k + \frac{\bar{\Delta}^2}{\alpha},$$

  where $\bar{\Delta} := \max_{u,v\in\{v_1,\ldots,v_M\}} \|u - v\|_2 \leq \sqrt{N}$.
- Consequently, within any stationary routing interval ($q^{(k+1)} = q^{(k)}$), tracking energy contracts geometrically as $E_{k+1} = (1-\alpha)^2 E_k < (1-\alpha)E_k$, while across switching epochs it is globally bounded by $\limsup_{k\to\infty} E_k \leq \bar{\Delta}^2/\alpha^2$.

Reviewer 2 — Comment R2.2(c) (Mathematical Rigor in Lemma 2 / Supplement Section S-I and Proposition 3 / Supplement Section S-J: Circular Conservation Assumption, Reversed Minimum Inequality, and `AND` vs. `OR` Slack Condition)

Reviewer's Suggestion (Restated): The Reviewer identifies two critical flaws in the performance bridge proofs: (1) in Lemma 2 (Appendix I), total betweenness centrality $\sum_i \kappa_i$ is not constant when path lengths change, and part 1 previously assumed $\kappa_{\max}^{(m+1)} \le \kappa_{\max}^{(m)}$ circularly without stating an explicit condition preventing diverted traffic from overloading another node $j \neq u$; and (2) in Proposition 3 (Appendix J), the inequality $\min_{i\in V}(\mu_i/\kappa_i') \ge \mu_u/\kappa_u'$ had its inequality sign reversed, and the heterogeneous sufficient condition requires a joint `AND` condition across nodes rather than `OR`.

Our Humble Response: We are deeply indebted to Reviewer 2 for catching the circular reasoning in Appendix I and the reversed minimum inequality in Appendix J. We humbly acknowledge both errors and have completely reconstructed **Lemma 2 (Section III-E and Supplement Section S-I)** and **Proposition 3 (Section III-E and Supplement Section S-J)**:

1. **Non-Circular Peak-Shaving Certificate in Lemma 2 (Supplement Section S-I)**: We removed any assumption that $\sum_i \kappa_i$ is conserved or that $\kappa_{\max}$ decreases a priori. Instead, for a routing switch at block $m \to m+1$ with pre-switch peak node u ($\kappa_u^{(m)} = \kappa_{\max}^{(m)}$) and positive load shedding $\Delta_u^- := \kappa_u^{(m)} - \kappa_u^{(m+1)} \ge \delta > 0$, we define each non-peak node's **Route-Redistribution Budget** $B_j^{(m)} := (\kappa_j^{(m+1)} - \kappa_j^{(m)})_+ \le \sum_{(s,t)\in\mathcal{D}_{\text{in}}^{(m)}(j)} W_{st}$. Lemma 2 proves that whenever the diverted traffic satisfies the explicit **Non-Overloading Diversion Condition** $\kappa_j^{(m)} + B_j^{(m)} \le \kappa_{\max}^{(m)} - \delta$ for all $j \in V \setminus \{u\}$, the post-switch global peak strictly satisfies $\kappa_{\max}^{(m+1)} \le \kappa_{\max}^{(m)} - \delta$.

2. **Corrected Minimum Inequality and Three-Part Capacity Bridge in Proposition 3 (Supplement Section S-J)**: Since $\rho_{\max}(\kappa) = \min_{i\in V}(\mu_i/\kappa_i)$ is the minimum over all nodes, we corrected the inequality direction to:

$$\rho_{\max}(\kappa) = \min_{i\in V} \frac{\mu_i}{\kappa_i} \ge \frac{\min_{i\in V}\mu_i}{\max_{i\in V}\kappa_i} = \frac{\mu_{\min}}{\kappa_{\max}} =: L(\kappa).$$

   Building on this exact relation, **Proposition 3** proves three rigorous statements:

   - *Part (i) Unconditional Expansion of Robust Capacity Lower Bound*: For any heterogeneous $\{\mu_i\}$, reducing $\kappa'_{\max} < \kappa_{\max}$ strictly increases the worst-case guaranteed capacity lower bound $L(\kappa') = \mu_{\min}/\kappa'_{\max} > \mu_{\min}/\kappa_{\max} = L(\kappa)$.

   - *Part (ii) Exact Inverse-Peak Capacity Proportionality*: When node service rates are homogeneous ($\mu_i \equiv \mu$, as in Table 2 / Table II and Fig. 5), $\min_{i\in V}(\mu/\kappa_i) = \mu/\kappa_{\max}$ holds with exact equality, yielding $\rho_{\max}(\kappa')/\rho_{\max}(\kappa) = \kappa_{\max}/\kappa'_{\max} > 1$. Moreover, for arbitrary heterogeneous capacities $\mu_i > 0$ and demands $W_{st} \ge 0$, using our generalized capacity-normalized stress $\widetilde{\kappa}_i(S; W, \mu)$ in Eq. (2) yields the exact identity $\rho_{\max}(\widetilde{\kappa}) = \min_{i\in V}(1/\widetilde{\kappa}_i) = 1/\widetilde{\kappa}_{\max}$, so $\rho_{\max}(\widetilde{\kappa}')/\rho_{\max}(\widetilde{\kappa}) = \widetilde{\kappa}_{\max}/\widetilde{\kappa}'_{\max} > 1$ whenever $\widetilde{\kappa}'_{\max} < \widetilde{\kappa}_{\max}$.

   - *Part (iii) Joint (`AND`) Sufficient Slack Condition under Heterogeneous μ_i with Unweighted BC*: Let $\rho_0 := \rho_{\max}(\kappa)$. If the relieved peak u (with load reduction $d := \Delta_u^- \ge \delta > 0$) **and** every non-peak node $j \in V \setminus \{u\}$ jointly satisfy:

$$\frac{\mu_u}{\kappa_{\max} - d} \ge \rho_0 \quad \textbf{and} \quad \frac{\mu_j}{\kappa_j + B_j} \ge \rho_0, \quad \forall j \in V \setminus \{u\},$$

   then $\rho_{\max}(\kappa') = \min_{i\in V}(\mu_i/\kappa_i') \ge \rho_0$ (and strictly $\rho_{\max}(\kappa') > \rho_0$ when the inequalities are strict).

Reviewer 2 — Comment R2.3 (Betweenness Centrality Under Non-Uniform Traffic Demands, Heterogeneous Node Capacities, and Time-Varying Surges)

Reviewer's Suggestion (Restated): The Reviewer notes that standard topological betweenness centrality assumes uniform all-pairs traffic and identical node capacities, and recommends discussing Demand-Weighted Betweenness or Capacity-Normalized Stress and testing SRA under skewed, heterogeneous, and time-varying traffic conditions.

Our Humble Response: We thank Reviewer 2 for this insightful and constructive suggestion, which inspired one of the most important theoretical and empirical extensions in our revised manuscript:

1. **Generalized Demand- and Capacity-Weighted Structural Stress $\widetilde{\kappa}_i(S; W, \mu)$ (Section III-A, Eq. (2))**: We extended the feedback signal from unweighted topological betweenness centrality $\kappa_i(S)$ to the generalized stress:

$$\widetilde{\kappa}_i(S; W, \mu) := \frac{1}{\mu_i} \sum_{s \neq i \neq t \in V} W_{st} \, \frac{\sigma_{st}^{C(S)}(i)}{\sigma_{st}^{C(S)}},$$

   which weights each shortest-path fraction by the origin-destination traffic demand $W_{st} \geq 0$ and normalizes by the node's service capacity $\mu_i > 0$. Because W_{st} and μ_i are constants within a quasi-stationary control horizon, $\widetilde{\kappa}(S; W, \mu)$ remains piecewise constant over the exact same finite polyhedral partition of $X = [0, 1]^N$, preserving all convergence theorems while making $\widetilde{\kappa}_{\max}$ an exact inverse measure of heterogeneous network capacity (**Proposition 3(ii)**).

2. **Empirical Validation Under Skewed Pareto Gravity Demand, Heterogeneous Capacities, and a $2\times$ Non-Stationary Traffic Surge (Section IV-G and Fig. 11)**: Across 30 independent trials on BA networks ($N = 100$) with heavy-tailed Pareto gravity demands ($w_i \sim$ Pareto(1.5), $W_{st} \propto w_s w_t$), heterogeneous capacities, and a $2\times$ traffic surge, **Generalized Weighted SRA** reduces peak generalized stress $\max_i \widetilde{\kappa}_i$ by 55.6% (from 2364.8 to 1051.1), improves throughput by +20.28% over Dijkstra (69.34 ± 12.14 vs. 57.65 ± 8.97 pkts/step, PDR 81.54% vs. 67.79%), reduces P95 tail latency from 12.93 to 9.83 steps, and outperforms unweighted **Topology-Only SRA** (68.91 ± 12.04 pkts/step).

Reviewer 2 — Comment R2.4 (Restoring Missing Figures, Dwell-Time Sensitivity T_d, Ablation of $T_d = 1$ and $\alpha = 1$, and Scalability/Runtime Benchmarks)

Reviewer's Suggestion (Restated): The Reviewer points out that three figures discussed in the text (`ren_fig2_lyapunov.pdf`, `ren_fig3_dwell_time.pdf`, and `sensitivity_3d_surfaces_full.pdf`) were missing or commented out, and requests restoring them alongside sensitivity analysis for T_d, ablations for $T_d = 1$ and $\alpha = 1$, and runtime/scalability benchmarks from $N = 50$ to 500 and on real backbone topologies.

Our Humble Response: We sincerely apologize to Reviewer 2 for the accidentally commented-out figures in our initial submission and thank you for your thorough check. We have resolved every item in full:

1. **Restored All Three Visual Figures**: We restored `ren_fig2_lyapunov.pdf` as **Fig. 6** (Section IV-B, Lyapunov tracking energy decay and switching spikes), `ren_fig3_dwell_time.pdf`

as **Fig. 7** (Section IV-B, phase-portrait trajectory stabilization and chattering suppression under $T_d = 5$ vs. $T_d = 1$), and `sensitivity_3d_surfaces_full.pdf` as **Fig. 12** (Section IV-H, joint 3D sensitivity surface over (α, β_I)).

2. **Ablation of $\alpha = 1$ and $T_d = 1$ & Sensitivity to $T_d \in \{1, 2, 3, 5, 8, 10\}$**: Added in **Section IV-F (Table 4 / Table IV, Fig. 10)** and **Section IV-H (Fig. 13(a))** across 30 trials, quantifying both the catastrophic throughput collapse at $\alpha = 1$ (53.39 pkts/step) and the 79.7% reduction in route switching frequency from $T_d = 1$ (98.3%) to $T_d = 5$ (20.0%).

3. **Scalability ($N = 50..500$), Real Carrier/AS Backbones (`Internet AS`, `Deltacom`, `UsCarrier`), and Runtime Overhead**: Added in **Section IV-H (Table 5 / Table V and Fig. 13(b))**, reporting wall-clock runtime (8.52 ms/eval on `Internet AS`, 36.96 ms/eval on `Deltacom`, 77.33 ms/eval on `UsCarrier`, 32.12 ms/eval at $N = 100$, and 902.95 ms/eval at $N = 500$) and demonstrating that SRA improves throughput by +18.91% **to** +81.07% on real carrier backbones and up to +108.33% on $N = 500$ scale-free graphs.

Reviewer 2 — Comment R2.5 (Minor Presentation and Parameter Clarification Issues: Stray Citation Artifact and Buffer Capacity vs. Service Rate in Table II)

*Reviewer's Suggestion (Restated): The Reviewer notes a stray text artifact '[cite: 276, 277]' in the caption of the congestion transition figure (originally Fig. 5) and asks the authors to clarify the distinction between Node Buffer Capacity (*50 *packets) and Node Service Rate (μ) in Table II.*

Our Humble Response: We thank Reviewer 2 for catching both details and humbly apologize for the oversight. We have:

1. Removed the stray '[cite: 276, 277]' text artifact from the caption of the congestion transition figure (now **Fig. 8** in the revised manuscript) and placed all literature citations properly in the main text;

2. Updated **Table 2 (Table II)** to clearly distinguish **Node Service Rate ($\mu = 1.0$ normalized steady-state budget; packet processing capacity** 10 **packets/timestep)** and **Queue Buffer Capacity ($B_{\max} = 50$ packets)** as explicit, unambiguous rows with units.

Once again, we would like to express our sincere appreciation to the Associate Editor-in-Chief, the Associate Editor, and both anonymous Reviewers for your invaluable guidance, patience, and constructive criticism, which have greatly improved our manuscript.